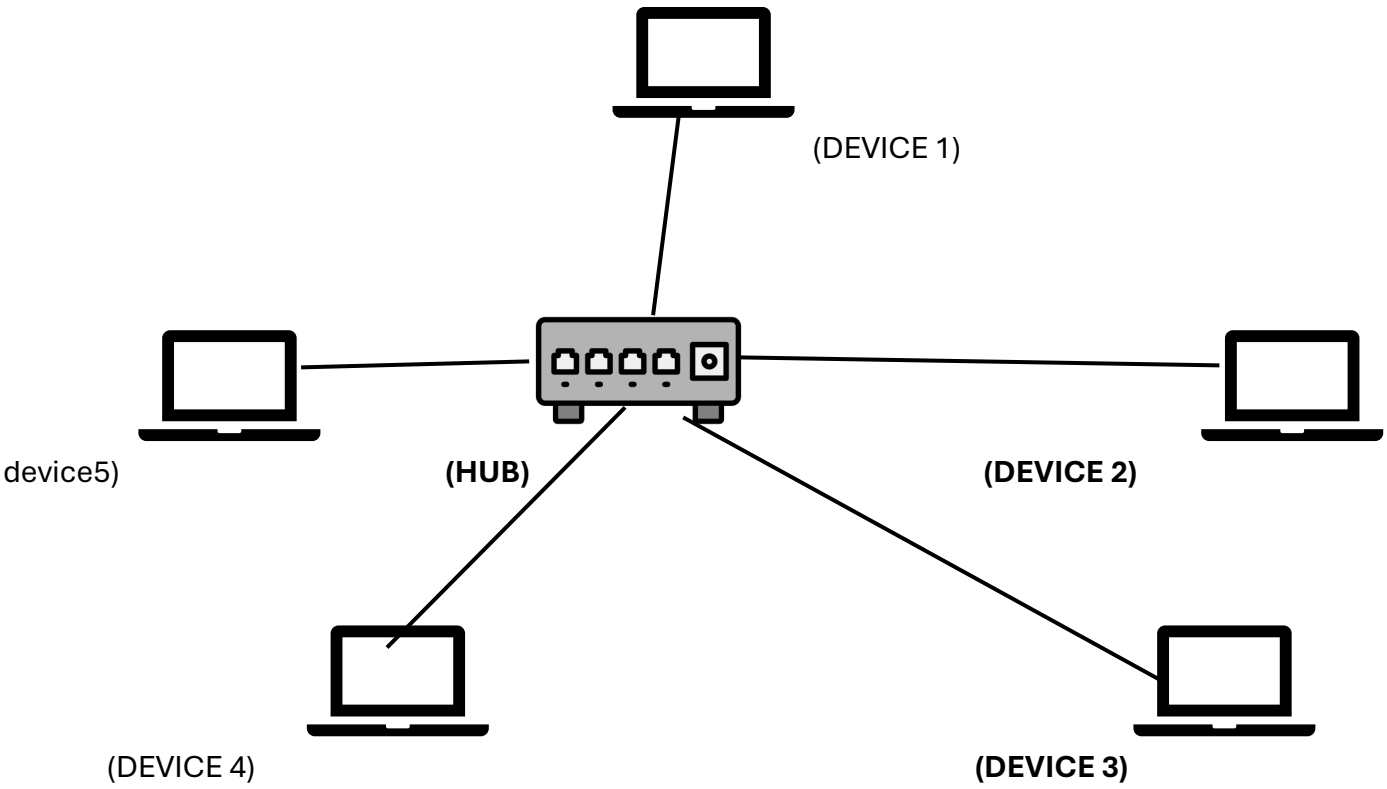
1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

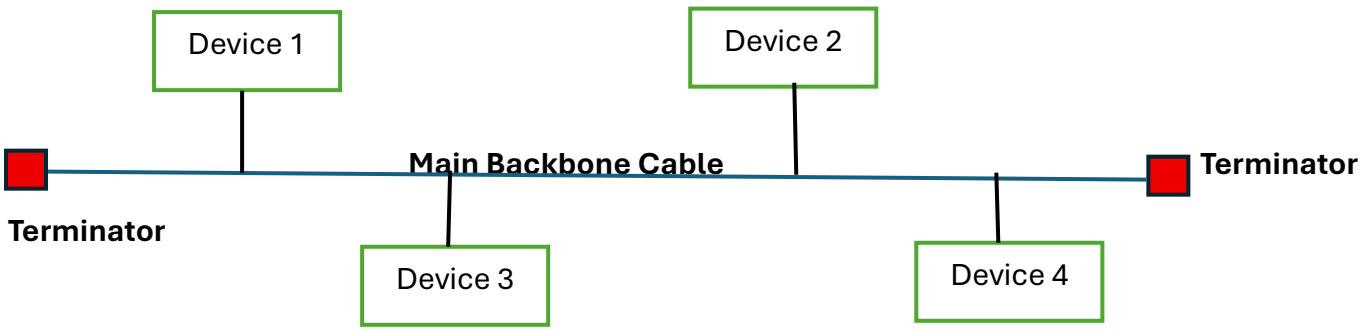
STAR TOPOLOGY



Real-World Service: Flipkart or any other shopping website

Why: All devices connect directly to Flipkart's central server to view products and make purchases If one customer's phone stops working, it does not stop anyone else from shopping.

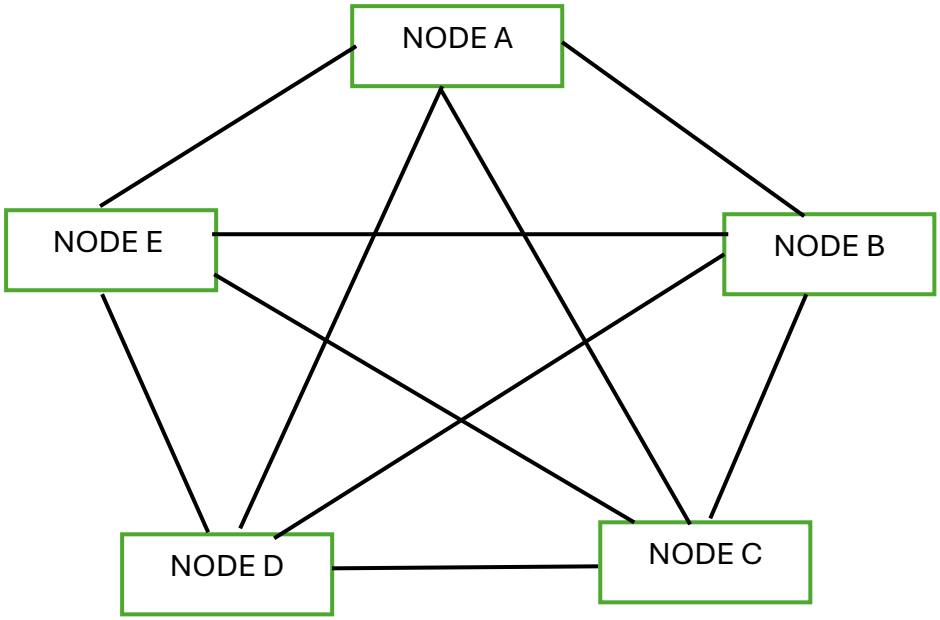
BUS TOPOLOGY



Real world example: IPL live broadcasting

Why: The main live video feed travels down to the single path (the backbone), and millions of viewers with just a tap receives the exact data stream at the same time.

MESH TOPOLOGY



Real world example: WhatsApp (Peer-to-Peer Calls or Encrypted Group Chats).

Why: During a call or chat, devices need a highly secure, direct, and reliable connection with each other. If one network route slows down, WhatsApp instantly reroutes your data through another path so your call doesn't drop.

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros	Cons	Best Use Case	App Example Scenario
Star	**Instagram Feed & Content Delivery:** When you open Instagram, your phone (the client node) requests images and videos directly from Instagram's central Cloud Data Centers/CDNs (the central hub).	**Instagram Feed & Content Delivery:** When you open Instagram, your phone (the client node) requests images and videos directly from Instagram's central Cloud Data Centers/CDNs (the central hub).	**Instagram Feed & Content Delivery:** When you open Instagram, your phone (the client node) requests images and videos directly from Instagram's central Cloud Data Centers/CDNs (the central hub).	**Instagram Feed & Content Delivery:** When you open Instagram, your phone (the client node) requests images and videos directly from Instagram's central Cloud Data Centers/CDNs (the central hub).

Bus	• Cheap and cost-effective to implement. • Uses minimal cabling. • Simple to set up for small networks. • No central switches required.	• Single cable failure breaks the whole network. • Heavy traffic causes major data collisions. • Difficult to isolate network faults. • Limits performance as more nodes are added.	Legacy systems or lightweight telemetry: Simple, sequential data transfer where high speed and uptime are not critical.	Zomato Delivery Executive Tracking (Internal Device Node): Very basic, low-power internal hardware or diagnostic testing benches where multiple sensors feed telemetry data into a single shared communication line.
Ring	• No data collisions (data moves in one direction). • High throughput under heavy load. • Equal network access for all nodes. • No central server needed to manage traffic.	• A single broken link breaks the entire loop. • Highly disruptive to add or remove nodes. • Slower performance than Star for large networks. • Harder to troubleshoot specific line faults.	High-speed token networks: Closed-loop industrial synchronization or high-speed localized data backbones.	Paytm Internal Batch Reconciliation (Dual-Ring): Closed internal banking database clusters that sync transaction ledger logs sequentially across redundant nodes in a highly predictable loop.
Mesh	• Extremely fault-tolerant with multiple backup paths. • No single point of failure. • Excellent security and privacy. • Handles massive traffic concurrent loads.	• Extremely expensive due to massive cabling/routing requirements. • Highly complex to configure and manage. • Redundant connections waste high bandwidth. • Massive hardware overhead.	Mission-Critical Systems: Inter-datacenter replication where zero downtime and constant availability are mandatory.	Paytm Payment Gateway & Core Banking: Paytm's core database nodes use a full mesh network. If a server rack processing UPI payments fails, the traffic instantly routes through another node to ensure your money transfer never fails.

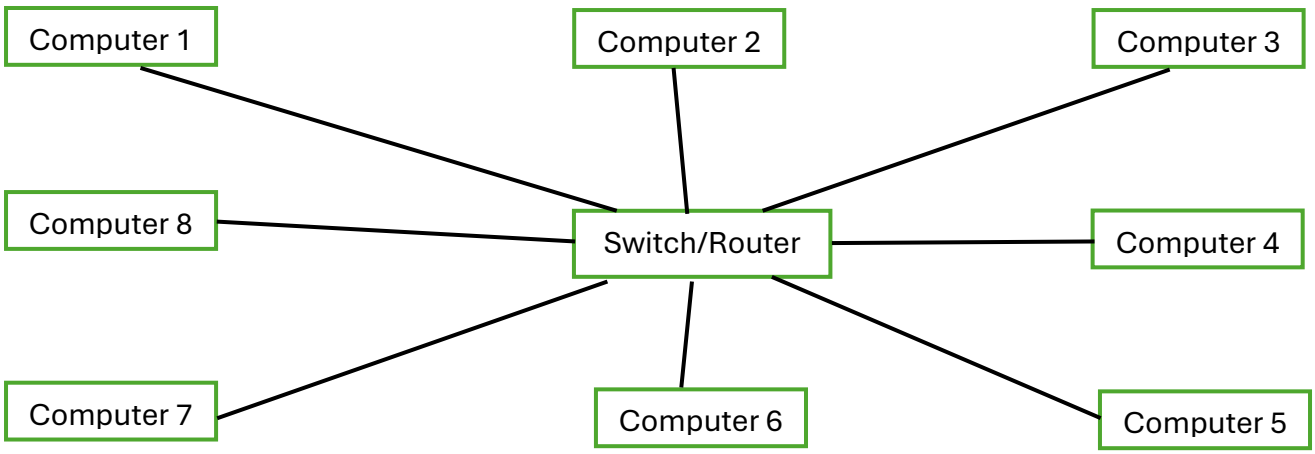
3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

For a multiplayer mobile game like PUBG or Free Fire, a Dedicated Client-Server (Hybrid) network topology is the ideal choice, utilizing localized regional edge servers.

1. **Speed:** It minimizes latency (ping) by connecting players to the closest regional server, ensuring fast, real-time synchronization essential for fast-paced shooters.
2. **Reliability:** It prevents cheating (as the server is the authoritative state holder) and ensures that if one regional server fails, only a subset of players is affected rather than the entire global player base.
3. **Cost:** While hosting dedicated servers incurs high infrastructure costs, it optimizes bandwidth consumption and allows for dynamic scaling—spinning servers up or down based on regional player demand to manage expenses.

4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.



For a small warehouse, a **Star Topology** is ideal because it is highly affordable and easily expands as the business grows. If one computer on the packaging line goes offline, the remaining seven computers continue processing orders without interruption, making it incredibly simple to troubleshoot